10 · Using a Punnett square

The words of this track, as it says them. Bold brown is the examiner.

Using a Punnett square. Four marks.

Explain how a Punnett square is used to predict offspring. Work a cross of big B little b with big B little b, where brown eyes, big B, are dominant over blue, little b. Then work a codominant blood group cross: I A little i, crossed with I B little i.

Genetics problems are the biggest pile of marks in her Module 3 quizzes, and this is the tool for nearly all of them.

The hook: one parent across the top, one down the side, and one letter from each into every box.

What it shows. Genes come in pairs of alleles, one from each parent. A Punnett square shows all the possible combinations of the parents' alleles at fertilisation, so it predicts the probability of each genotype in the offspring.

How to draw it. One parent's two alleles go across the top. The other parent's two alleles go down the side. Fill each of the four boxes with one allele from the top and one from the side.

Cross one: big B little b, crossed with big B little b. The boxes: one big B big B, two big B little b, and one little b little b. One to two to one. As percentages, that's twenty-five per cent homozygous dominant, fifty per cent heterozygous, and twenty-five per cent homozygous recessive. Now the phenotype. Big B big B and big B little b both show the dominant trait, so it's seventy-five per cent brown, twenty-five per cent blue. Only little b little b shows blue.

Cross two, the blood groups. This is codominance: I A and I B are both expressed, and little i, which gives group O, is recessive. I A little i, crossed with I B little i. The four boxes: I A I B, which is group A B. I A little i, group A. I B little i, group B. And little i little i, group O. Twenty-five per cent each. That's how four children can have four different blood groups.

The traps. Genotype is the letters. Phenotype is what shows: brown or blue, or the blood group. When she asks for both, give both. Second: each box is a probability for each child, not a count of real children. Third, the words: heterozygous is two different letters, big B little b. Homozygous is two the same.

Once more. The Punnett square, and both crosses.

Alleles come in pairs, one from each parent, and the Punnett square shows every possible combination, so the probability of each. One parent across the top, the other down the side. Big B little b crossed with big B little b gives one to two to one: twenty-five per cent homozygous dominant, fifty heterozygous, twenty-five homozygous recessive. So seventy-five per cent brown, twenty-five blue. In codominance, I A and I B are both expressed: I A little i crossed with I B little i gives group A B, group A, group B and group O, twenty-five per cent each.